AI-Powered Collections Strategy

Leveraging Agentic AI for Scalable, Fair, and Effective Debt Management at Geldium

Executive briefing for Geldium's Head of Collections | Helga Yuliani Putri Aritonang, Tata iQ | October 2026

How the System Works: From Data to Action

1 INPUTS
Customer data

- Payment activity: due dates, late and missed payments
- Credit use and debt-to-income
- Account tenure and product
- Contact history and customer replies

2 DECISION LOGIC
Decision engine

- Assigns a risk tier using transparent rules, or the model once it passes the readiness gate
- Chooses the next best action, timing and channel
- Checks every action against guardrails

3 ACTIONS
Action layer

- Reminders by SMS, email or app
- Self-service repayment-plan offers
- Hand-off to a collections officer for sensitive cases

4 LEARNING
Learning loop

- Compares outcomes with a control group
- Learns which message, time and channel work
- Monthly model and fairness re-tests

Built-in data check: records with conflicting payment fields (28% of today's sample) are held for correction instead of driving decisions. **Age and location are never used to decide actions.**

Decision Logic: Risk Tiers and Next Best Actions

Risk tier	Trigger (Phase 1 rules)	What the AI agent does	Human role
Low	Debt-to-income ≤ 20% or credit use ≤ 30%, payments up to date	Standard due-date reminder	None needed
Medium	Debt-to-income > 20% and credit use > 30%	Reminder 5 days before due date plus self-service plan link; adapts timing and channel to responses	Monitors dashboard
High	Medium criteria plus account ≤ 2 years old, or a first missed payment	Priority outreach and an offer to discuss a repayment plan	Approves any plan terms
Critical	30+ days past due, a hardship signal, or a dispute	Acknowledges the customer and hands the case to a person	Collections officer owns the case

Readiness gate: rules first, model second

Today's prediction model performs no better than chance, so Phase 1 runs on the transparent rules above. Model scores replace rules only once they reach AUC ≥ 0.70 and catch at least 70% of delinquent customers on fresh data, and pass the fairness tests.

Role of Agentic AI: Autonomous vs Human Oversight

Autonomous

The AI agent decides and acts

- Sends payment reminders and confirmations
- Chooses timing and channel within contact limits
- Adjusts cadence: pauses when a customer engages, steps up when they don't
- Offers self-service options: pay now, autopay, request a plan
- Routes cases to the right human queue

Human oversight

The agent recommends, a person decides

- Approves hardship plans and payment deferrals
- Handles disputes, complaints and vulnerable customers (illness, job loss)
- Reviews cases the agent flags as uncertain or outside the rules
- Approves every change to rules, targets or the model

Never automated (locked): raising interest rates or fees, cutting credit limits, declining hardship support, and referral to external collectors or legal action. **Only a person can take these steps.**

Responsible AI Guardrails

1 Fairness

- Age, location and age-linked fields (e.g., "Student" card) excluded from decisions
- Monthly disparate-impact test: any group treated at < 80% the rate of another pauses that strategy

2 Explainability

- Every action logged with a plain-language reason, e.g., "balance high relative to income"
- Reason codes (SHAP) once the model is live; customers can ask why and appeal

3 Compliance

- Designed to ECOA and FCRA standards: no discrimination, accurate data
- Contact caps modelled on CFPB Regulation F: ≤ 7 call attempts in 7 days, easy SMS/email opt-out

4 Human control

- Locked-action list, kill switch and full audit trail
- Compliance signs off every rule change; quarterly independent review

Expected Business Impact

79%

of today's delinquent accounts sit in the segment the system contacts first (Medium tier and above).

≥ 10%

target cut in 30+ day delinquency vs the control group, measured in the pilot before scaling.

Business KPIs

Targets to validate in the pilot

- 30+ day delinquency rate: down ≥ 10% vs control
- Cure rate: more late accounts back to current within 30 days
- Roll rate: fewer accounts moving from 30 to 60+ days late
- Cost to collect: lower, as routine contact is automated
- Agent time shifted to high-risk and hardship cases

Customer outcomes

Tracked alongside every KPI

- Earlier, supportive contact before a payment is missed
- Fewer, better-timed messages within contact caps
- Clear reasons for every action and a route to appeal
- Consistent treatment across age and location groups
- Scales to the whole book without matching headcount

Rollout Roadmap and Decision Needed

0 — 1 — 2 — 3

Phase 0 | Weeks 0–4

Fix the data

Correct delinquency and payment-history fields; standardise labels

Exit: conflicting records resolved

Phase 1 | Weeks 2–10

Rules-based pilot

Agent runs Low–High tiers on rules, with a 50/50 control group

Exit: ≥ 10% cut vs control and fairness tests pass

Phase 2 | After the gate

Model-driven

Model scores replace rules once AUC ≥ 0.70 and fairness tests pass

Exit: model beats rules in a head-to-head test

Phase 3 | Ongoing

Scale and learn

Roll out to the full book; monthly re-tests and quarterly audits

Kill switch and audits stay on permanently

Decision needed today: approve Phase 0 and Phase 1. Both are low cost, use existing channels and staff, and give Geldium hard evidence before any wider automation.